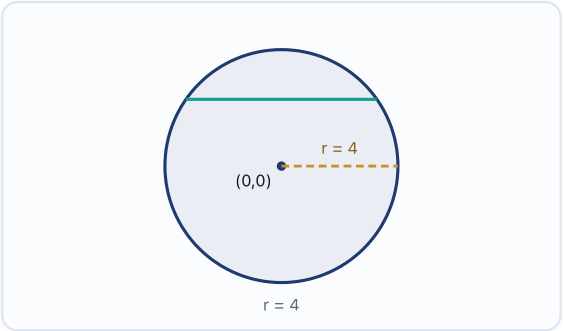
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Trigonometric Functions & Circular Measure Practice Exercises

1 MCQ

A sector of a circle of radius $r = 8$ cm has central angle $\theta = 0.9$ rad. What is the area of the sector?



A $28.8\ \text{cm}^2$

B $57.6\ \text{cm}^2$

C $33.8\ \text{cm}^2$

D $7.2\ \text{cm}^2$

2 MCQ

A sector of a circle of radius $r = 10$ cm has central angle $\theta = 1.0$ rad. What is the area of the sector?

A $50.0\ \text{cm}^2$

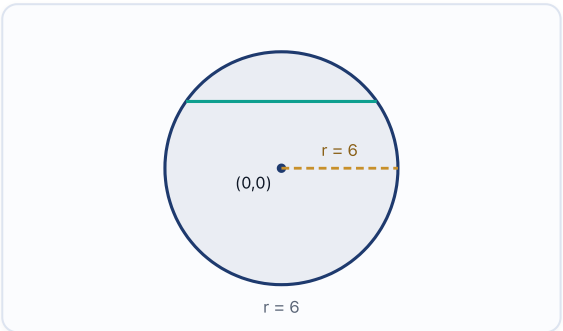
B $100.0\ \text{cm}^2$

C $55.0\ \text{cm}^2$

D $10.0\ \text{cm}^2$

3 MCQ

A sector of a circle of radius $r = 12$ cm has central angle $\theta = 1.1$ rad. What is the area of the sector?



A $79.2\ \text{cm}^2$

B $158.4\ \text{cm}^2$

C $84.2\ \text{cm}^2$

D $13.2\ \text{cm}^2$

4 MCQ

A sector of a circle of radius $r = 14$ cm has central angle $\theta = 1.2$ rad. What is the area of the sector?

A $16.8\ \text{cm}^2$

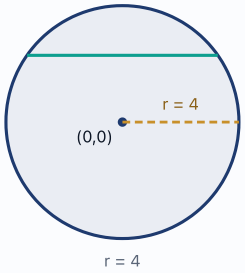
B $117.6\ \text{cm}^2$

C $235.2\ \text{cm}^2$

D $122.6\ \text{cm}^2$

5 MCQ

A sector of a circle of radius $r = 16$ cm has central angle $\theta = 1.3$ rad. What is the area of the sector?



A $332.8\ \text{cm}^2$

B $20.8\ \text{cm}^2$

C $166.4\ \text{cm}^2$

D $171.4\ \text{cm}^2$

6 MCQ

A sector of a circle of radius $r = 18$ cm has central angle $\theta = 1.4$ rad. What is the area of the sector?

A $453.6\ \text{cm}^2$

B $25.2\ \text{cm}^2$

C $231.8\ \text{cm}^2$

D $226.8\ \text{cm}^2$

7 MCQ

A sector of a circle of radius $r = 20$ cm has central angle $\theta = 1.5$ rad. What is the area of the sector?

A $300.0\ \text{cm}^2$

B $600.0\ \text{cm}^2$

C $305.0\ \text{cm}^2$

D $30.0\ \text{cm}^2$

8 MCQ

Find all solutions of $2\cos^2 x + 3\sin x - 3 = 0$ in the interval $0 \le x \le 2\pi$.

A $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{\pi}{2}$

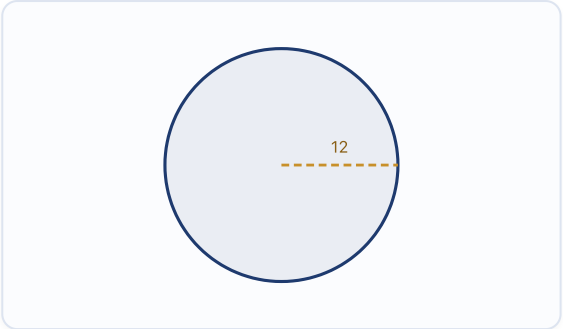
B $x = \frac{\pi}{3},\ \frac{2\pi}{3}$

C $x = \frac{\pi}{6},\ \frac{7\pi}{6}$

D $x = 0,\ \pi$

9 MCQ

In an applied problem, a sector of a circle of radius $r = 12$ cm has central angle $\theta = 1.1$ rad. What is the area of the sector?



A $158.4\ \text{cm}^2$

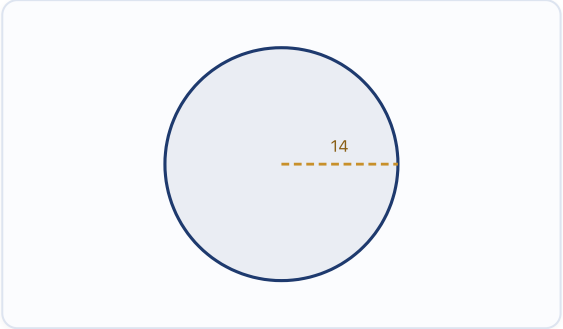
B $84.2\ \text{cm}^2$

C $13.2\ \text{cm}^2$

D $79.2\ \text{cm}^2$

10 MCQ

Suppose a sector of a circle of radius $r = 14$ cm has central angle $\theta = 1.2$ rad. What is the area of the sector?



A $16.8\ \text{cm}^2$

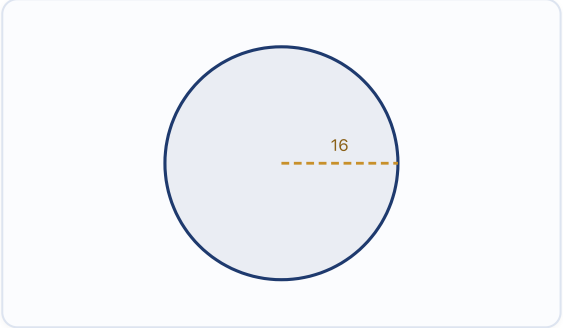
B $235.2\ \text{cm}^2$

C $117.6\ \text{cm}^2$

D $122.6\ \text{cm}^2$

11 MCQ

An observed sector of a circle of radius $r = 16$ cm has central angle $\theta = 1.3$ rad. What is the area of the sector?



A $171.4\ \text{cm}^2$

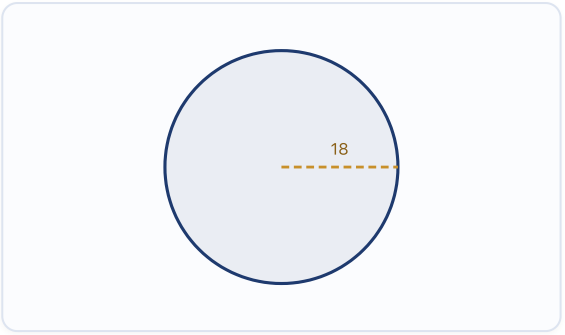
B $332.8\ \text{cm}^2$

C $20.8\ \text{cm}^2$

D $166.4\ \text{cm}^2$

12 MCQ

Consider a sector of a circle of radius $r = 18$ cm has central angle $\theta = 1.4$ rad. What is the area of the sector?



A $453.6\ \text{cm}^2$

B $226.8\ \text{cm}^2$

C $25.2\ \text{cm}^2$

D $231.8\ \text{cm}^2$

13 MCQ

Find all solutions of $2\sin^2 x - \sin x - 1 = 0$ in the interval $0 \le x \le 2\pi$.

A $x = \frac{\pi}{2},\ \frac{7\pi}{6},\ \frac{11\pi}{6}$

B $x = \frac{3\pi}{2},\ \frac{\pi}{6}$

C $x = \frac{\pi}{6},\ \frac{5\pi}{6}$

D $x = \pi,\ 2\pi$

14 MCQ

Find all solutions of $2\cos^2 x + \cos x - 1 = 0$ in the interval $0 \le x \le 2\pi$.

A $x = \frac{\pi}{2},\ \frac{3\pi}{2}$

B $x = 0,\ \pi$

C $x = \pi,\ \frac{\pi}{3},\ \frac{5\pi}{3}$

D $x = \frac{2\pi}{3},\ \frac{4\pi}{3}$

15 MCQ

Find all solutions of $2\sin^2 x - 3\sin x + 1 = 0$ in the interval $0 \le x \le 2\pi$.

A $x = \frac{\pi}{3},\ \frac{2\pi}{3}$

B $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{\pi}{2}$

C $x = \frac{3\pi}{2}$

D $x = \frac{7\pi}{6},\ \frac{11\pi}{6}$

16 MCQ

Find all solutions of $2\cos^2 x - 3\cos x + 1 = 0$ in the interval $0 \le x \le 2\pi$.

A $x = 0,\ \frac{\pi}{3},\ \frac{5\pi}{3},\ 2\pi$

B $x = \frac{\pi}{2},\ \frac{3\pi}{2}$

C $x = \pi$

D $x = \frac{\pi}{6},\ \frac{11\pi}{6}$

17 MCQ

Find all solutions of $4\cos^2 x - 3 = 0$ in the interval $0 \le x \le 2\pi$.

A $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{7\pi}{6},\ \frac{11\pi}{6}$

B $x = \frac{\pi}{3},\ \frac{2\pi}{3},\ \frac{4\pi}{3},\ \frac{5\pi}{3}$

C $x = 0,\ \pi$

D $x = \frac{\pi}{4},\ \frac{3\pi}{4}$

18 MCQ

Find all solutions of $3\tan^2 x - 1 = 0$ in the interval $0 \le x \le 2\pi$.

A $x = \frac{\pi}{3},\ \frac{2\pi}{3},\ \frac{4\pi}{3},\ \frac{5\pi}{3}$

B $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{7\pi}{6},\ \frac{11\pi}{6}$

C $x = \frac{\pi}{2},\ \frac{3\pi}{2}$

D $x = \frac{\pi}{4},\ \frac{5\pi}{4}$

19 MCQ

In $\triangle ABC$, sides of lengths 32 and 10 meet at an included angle of 60° opposite side x. Find the squared length $f = x^2$ by the Law of Cosines.

A 803

B 820

C 804

D 809

20 MCQ

In triangle ABC, $AB = 6$ cm, $AC = 10$ cm, and the area of triangle ABC is $15\sqrt{3}$ cm^2. Given that angle BAC is obtuse, find the exact length of BC.

A 14 cm

B 8 cm

C $2\sqrt{19}$ cm

D $2\sqrt{34}$ cm

21 MCQ

A sector OAB of a circle with centre O and radius 12 cm has angle $\angle AOB = \frac{\pi}{2}$ radians. The point C is the midpoint of the radius OA. The line passing through C perpendicular to OA intersects the circular arc AB at the point D. The straight line segment BC is drawn.

Find the exact area, in cm^2, of the region bounded by the line segment CD, the circular arc BD, and the line segment BC.

A $12\pi + 18\sqrt{3} - 36$

B $12\pi + 36\sqrt{3} - 36$

C $18\pi + 18\sqrt{3} - 36$

D $24\pi + 18\sqrt{3} - 36$

22 MCQ

The function f is defined for all real numbers x by

$$\mathrm{f}(x) = 2 - 3\sin(4x).$$

What are the amplitude and the period (in radians) of f?

A Amplitude $= -3$, Period $= \frac{\pi}{2}$

B Amplitude $= 3$, Period $= 8\pi$

C Amplitude $= 5$, Period $= \frac{\pi}{4}$

D Amplitude $= 3$, Period $= \frac{\pi}{2}$

23 MCQ

In triangle ABC, the side lengths are $AB = (x+2)$ cm, $BC = (3x-1)$ cm, and $AC = 7$ cm. Given that $\angle ABC = 60^\circ$ and $x > 0$, find the exact area of triangle ABC.

A $10\sqrt{3}$ cm^2

B 10 cm^2

C $5\sqrt{3}$ cm^2

D $20\sqrt{3}$ cm^2

24 MCQ

Two circles, C_1 with centre O_1 and radius 6 cm, and C_2 with centre O_2 and radius $6\sqrt{3}$ cm, intersect at the points A and B. The distance between their centres is $O_1O_2 = 12$ cm.
Find the exact area of the region common to both circles, in cm^2.

A $30\pi + 36\sqrt{3}$

B $30\pi - 36\sqrt{3}$

C $42\pi - 36\sqrt{3}$

D $15\pi - 18\sqrt{3}$

25 MCQ

A sector of a circle of radius 12 cm has an arc length of 18 cm. Find the area of the sector.

A 216 cm^2

B 54 cm^2

C 108 cm^2

D 144 cm^2